THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two plane mirrors are placed at 90° to each other. A ray falls on one mirror at 30° to its surface. After two reflections the ray is: 1

(a) perpendicular to the incident ray
(b) absorbed
(c) parallel to the incident ray but opposite in direction
(d) along the same direction

2. During a solar eclipse, the Moon's shadow falls on the Earth. The region with total darkness is the: 1

(a) penumbra **(b)** umbra **(c)** halo **(d)** corona

3. Why does the image in a mirror appear to be as far behind as the object is in front? 1

(a) light bends
(b) virtual image position follows the law of reflection
(c) mirror is smaller
(d) mirror is thick

4. A ray is incident on a plane mirror at 25° to the normal. If the mirror is turned so that the angle of incidence becomes 35°, the reflected ray turns by: 1

(a) 20° **(b)** 35° **(c)** 10° **(d)** 60°

5. During a lunar eclipse, the Moon passes into Earth's: 1

(a) orbit only **(b)** shadow **(c)** atmosphere only **(d)** light

6. A mirror reflects the colours of objects. A red rose in white light viewed in a plane mirror appears: 1

(a) green **(b)** black **(c)** red **(d)** white

7. A person 1.6 m tall wishes to see a full-length view in a plane mirror. The minimum mirror length is: 1

(a) 1.6 m **(b)** 0.4 m **(c)** 3.2 m **(d)** 0.8 m

8. A student sees a shadow of her hand on a screen 2 m from a lamp. If she moves the hand to the midpoint between lamp and screen, the shadow size relative to the hand is: 1

(a) same **(b)** half **(c)** four times **(d)** double

9. The colour of the shadow of a red ball in white light is: 1

(a) red **(b)** white **(c)** black **(d)** green

10. A girl walks away from a mirror at 2 m/s for 3 s. How far has the distance between her and her image increased? 1

(a) 12 m **(b)** 2 m **(c)** 6 m **(d)** 3 m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A ray is incident at 90° to a plane mirror. State the angle of incidence and the path of the reflected ray. 2

12. Why is the shadow of a tall pole longer in the evening than at noon? 2

13. Explain why the Moon looks bright at night but is not a luminous body. 2

14. A plane mirror forms an image of a 5 cm tall object. What are the image height and nature? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ray of light strikes a mirror with an incident angle of 55°. Find the angle between the incident ray and the reflected ray, and the angle between the mirror and the reflected ray. 3

16. Describe how to use a stick and its shadow to find the time of noon. 3

17. Why do you see your face in a steel spoon but not as clearly as in a mirror? 3

OR

A girl is 4 m from a mirror and walks 1 m towards it. By how much does the distance between her and her image decrease? 3

18. Why are reflecting surfaces on roads and bicycles used? 3

OR

Explain why a mirror placed in a dark room cannot be seen. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two plane mirrors are inclined at 60° with an object between them. Find the number of images, explain how they form, and describe how you could verify it. 5

OR

Compare a pinhole camera and a plane mirror as image forming devices. 5

20. Design an experiment to find the relation between the object distance and the image distance in a plane mirror, with a table and conclusion. 5

OR

Explain how a plane mirror in a barber's shop shows the back of a customer's head, using a second mirror. 5

END OF QUESTION PAPER — Total: 40 Marks